

Rupesh Bodkhe

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PROFILE SUMMARY

AI and Backend Engineer with ~4 years of experience building and shipping end-to-end backend and AgenticAI applications using diverse Python frameworks. Experienced in rapidly developing microservices, event-driven architectures, and multi-agent & GenAI systems on cloud-native infrastructure. Comfortable owning features from design to production, integrating LLMs, APIs, frontend components, and DevOps pipelines to deliver scalable products.

SKILLS

Programming Languages	: Python, SQL (Postgres, Oracle)
GenAI	: LLMs (like OpenAI & Claude), Azure AI Foundry, AWS Bedrock, NLP, Prompt Engineering, Vector Databases, LangChain, RAG
AgenticAI	: MultiAgent Architecture, Context Engineering, DeepAgents, LangGraph, MCP, A2A Protocol, Google ADK
Backend & APIs	: Django REST Framework, FastAPI, Celery, Redis
Frontend	: React.js
Cloud & DevOps	: AWS, Azure, GCP, Docker, Github, Github Actions, Jenkins, Nginx
Software Engineering skills	: API Design and Development, Data Structures and Algorithms, System Design, Microservices and Event-based architectures, SDLC, AI Tools
Data	: Pandas, Numpy, Airflow

WORK EXPERIENCE

HASHEDIN BY DELOITTE | Software Engineer 2 (AI & Backend)

Pune, India | Jan 2025 – Present

Project 1: GenAI and AgenticAI SaaS Applications

- Worked on developing 3 enterprise GenAI & Agentic applications: ITSM Agent, Multi-Agent System, Data Analytics Agent.
- Authored ~75,000 tokens of production-grade natural language prompts for diverse GenAI tasks (retrieval, evaluation, reasoning, and summarization, etc) to significantly reduce LLM hallucinations and improve response accuracy from 50% to 99% in a Fortune 500 facing ITSM SaaS application.
- Lead developer of RAG microservice, Knowledge Article Generator microservice (cloud agnostic asynchronous event based), Prompt Engineering, MultiAgent System and Data Analytics Agent.
- Designed and productionized a centralized MultiAgent Orchestrator System with advanced Context Engineering leveraging Google-ADK, A2A and MCP protocols on a FastAPI backend.
- Led the design and development of Data Analytics Assistant from scratch to production in record 1 month, enabling analysts to generate SQL results, CSV files, and plotting graphs, significantly reducing repetitive manual work.
- Architected and developed a cloud-agnostic microservice by migrating an existing Flask and Airflow DAG using asynchronous event based architecture centered on FastAPI, Celery, Redis and PostgreSQL.

Project 2: Enterprise web app with LLM based chatbot

- Designed, developed, and deployed two auto-scaling LLM-powered RAG chatbot web applications, serving internal leadership and over 400,000 global employees.
- Leveraged Anthropic Claude and OpenAI LLMs to build secure, context-aware conversational systems using RAG grounded exclusively in organizational data.
- Hosted the Django backend on AWS App Runner to leverage serverless architecture for efficient deployment, cost optimization and auto-scaling.
- Collaborated cross-functionally with project managers, frontend developers, QA, and DevOps teams to define APIs, align chatbot behavior expectations, and ensure smooth scaling and deployment pipelines.

CLSA | Software Engineer (Fullstack)

Pune, India | July 2022 – Dec 2024

- Developed a high-performance ETL system with Python, Pandas, and Oracle SQL, reducing data processing time from 8 hours to under 5 minutes (>98% improvement) and significantly accelerating BAU operations.
- Designed, developed, and deployed two full-stack applications (monitoring & scheduling platform and file-sharing system) from scratch to production, improving team productivity and operational efficiency.
- Automated ETL pipelines for data extraction from Calypso to regulatory repositories (HKTR, MAS), eliminating manual intervention and ensuring 100% compliance with global reporting standards
- Led a team of developers, guided three new members and two graduates, and additionally served as a code reviewer.

EDUCATION

SAVITRIBAI PHULE PUNE UNIVERSITY | Bachelor of Engineering - Computer Science

Pune, India | Aug 2018 - May 2022

- CGPA: 8.91/10
- Honors in Artificial Intelligence and Machine Learning (2 years)

CERTIFICATIONS & ACHIEVEMENTS

- Certification (from Amazon): [AWS Certified Developer - Associate](#)
- Certification (from Amazon): [AWS Partner: Accreditation \(Technical\) & AWS Technical Essentials](#)
- Certification (from Google): [Google Cloud Certified Professional Cloud DevOps Engineer](#)
- Certification (from Anthropic): [Anthropic Claude with Amazon Bedrock](#)
- Certification (from Hashedin By Deloitte): GenAI Essential Training
- **Awarded** with the Star Award in recognition of exceptional performance and **innovative projects** for my current organization - CLSA.
- **Awarded** with the Spot Award for making an impact by **leading the development of a GenAI project** and delivering it on time - Hashedin by Deloitte.
- **Awarded** with the Spot Award for **GenAI based SaaS project** for my significant impact in improving accuracy of entire GenAI application from **50% to upto 99% using prompt engineering** - Hashedin by Deloitte.
- **Awarded** with the **Top Impactor Award** for **high-impact contributions**, timely delivery, and leadership across frontend, backend, DevOps and GenAI initiatives.

PROJECTS

Enterprise Chatbot & API Service

Technologies: Python, Django, MSAL, JWT, AWS Bedrock (Claude), OpenAI GPT, LangChain, AWS App Runner, PostgreSQL, Embeddings, Prompt Engineering, FAISS, NLP

- Developed a secure, production-grade enterprise web platform with RAG-based chatbots (AWS Bedrock, OpenAI, FAISS, FastAPI), deployed on an auto-scaling cloud architecture to enable context-aware knowledge retrieval grounded in organizational data.

Multi-Agent System Orchestrator Microservice

Technologies: A2A Protocol, MCP Protocol, LLMs, Python, FastAPI, Context Engineering, Prompt Engineering

- Designed and implemented a multi-agent system, initially as a POC and subsequently productionized as a standalone microservice, capable of orchestrating hundreds of MCP tools and remote A2A compatible Agents.

Data Analytics Chatbot

Technologies: Python, FastAPI, LangChain Agent, Prompt Engineering Plotly, Pandas, AWS S3

- Built a fully functional enterprise data analytics chatbot within a month, helping data analysts with day-to-day work generating complex SQL queries, converting data to csv and generating graphs, significantly reducing manual work.

PUBLICATIONS

- **Medium Article:** Building a Multi-Agent System with Google ADK: A Deep Dive into the MultiAgent Project ([Link](#))